

Batch Gradient Descent — Derivation of the Gradient Formulas

1. The Linear Regression Model

Suppose our simple linear regression model is:

$$y_{\text{pred}} = mx + b$$

m = slope **b** = intercept **x** = input **y** = actual value **n** = number of observations

2. Start with the Error

For observation *i*:

$$e_i = y_i - y_{\text{pred},i}$$

Since $y_{\text{pred},i} = mx_i + b$:

$$e_i = y_i - (mx_i + b)$$

3. Define the MSE Cost Function

Batch Gradient Descent minimizes the Mean Squared Error (MSE):

$$J(m,b) = (1/n) \sum [y_i - y_{\text{pred},i}]^2$$

or equivalently:

$$J(m,b) = (1/n) \sum [y_i - (mx_i + b)]^2$$

4. Derive gradient_m

We differentiate *J* with respect to *m*:

$$\partial J / \partial m = (1/n) \sum 2(y - y_{\text{pred}}) \cdot \partial(y - y_{\text{pred}}) / \partial m$$

Because $y_{\text{pred}} = mx + b$:

$$\partial y_{\text{pred}} / \partial m = x$$

Therefore:

$$\partial(y - y_{\text{pred}}) / \partial m = -x$$

Substituting:

$$\partial J / \partial m = (1/n) \sum 2(y - y_{\text{pred}})(-x)$$

$$\partial J / \partial m = -(2/n) \sum x(y - y_{\text{pred}})$$

This gives the Python formula:

$$\text{gradient_m} = -(2/n) * \text{sum}(x * (y - y_{\text{pred}}))$$

5. Derive gradient_b

Again, differentiate J with respect to b:

$$\partial J / \partial b = (1/n) \sum 2(y - y_{\text{pred}}) \cdot \partial(y - y_{\text{pred}}) / \partial b$$

Since $y_{\text{pred}} = mx + b$:

$$\partial y_{\text{pred}} / \partial b = 1$$

Therefore:

$$\partial(y - y_{\text{pred}}) / \partial b = -1$$

Substituting:

$$\partial J / \partial b = (1/n) \sum 2(y - y_{\text{pred}})(-1)$$

$$\partial J / \partial b = -(2/n) \sum (y - y_{\text{pred}})$$

This gives the Python formula:

$$\text{gradient_b} = -(2/n) * \text{sum}(y - y_{\text{pred}})$$

6. Gradient Descent Updates

Once the gradients are calculated, m and b are updated using the learning rate α :

$$m_{\text{new}} = m_{\text{old}} - \alpha(\partial J / \partial m)$$

$$b_{\text{new}} = b_{\text{old}} - \alpha(\partial J / \partial b)$$

In Python:

$$\begin{aligned} m &= m - \text{learning_rate} * \text{gradient_m} \\ b &= b - \text{learning_rate} * \text{gradient_b} \end{aligned}$$

7. The Complete Logic

$$y_{\text{pred}} = mx + b$$

↓

$$J = (1/n) \sum (y - y_{\text{pred}})^2$$

↓ differentiate

$$\begin{aligned} \text{gradient_m} &= -(2/n) \sum x(y - y_{\text{pred}}) \\ \text{gradient_b} &= -(2/n) \sum (y - y_{\text{pred}}) \end{aligned}$$

↓

Update m and b



Repeat until the error becomes small.

8. The Most Important Point

The minus sign comes from the expression $y - y_{\text{pred}}$. Because $y_{\text{pred}} = mx + b$, differentiating the negative prediction produces the minus sign. This is what makes the gradient point in the direction of increasing error, so Gradient Descent moves in the opposite direction to reduce the error.